

Images on the **Web**

HTML course: Lesson 5



Lesson Plan

1

Different image formats

2

Methods of inserting images into a website

3

Working with decorative properties

4

Optimization

Formats



Image Formats

Raster Images

- Consist of pixels
- Have a fixed resolution

- JPEG
- PNG
- GIF
- **WEBP**
- AVIF



1x



4x

Vector Images:

- Created using geometric **shapes**, such as **lines and curves**
- Can be **changed in CSS**

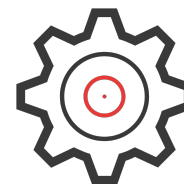
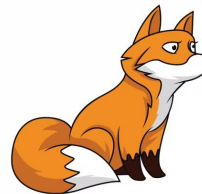
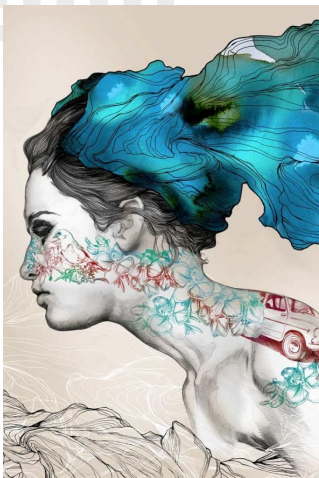
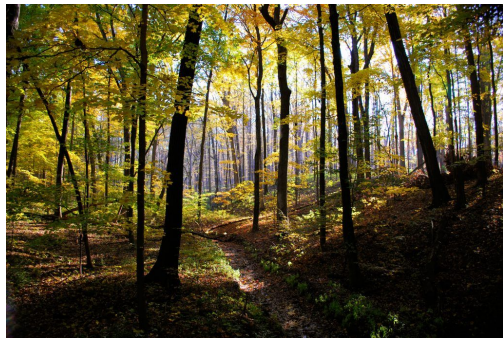
- SVG



1x



4x



Anatomy of SVG

<circle>

<rect>

<path>

<ellipse>

<polygon>

<text>

Grouping container – <g>

Hidden element – <defs>

Reuse an element – <use>

```
1 <div class="h-screen flex justify-center items-center">
2   <svg
3     xmlns="http://www.w3.org/2000/svg"
4     fill="none"
5     viewBox="0 0 24 24"
6     stroke-width="1"
7     stroke="rebeccapurple"
8     class="h-72 w-72"
9   >
10    <path
11      stroke-linecap="round"
12      stroke-linejoin="round"
13      d="M21 8.25c0-2.485-2.099-4.5-4.688-4.5-1.935 0-3.597
14        1.126-4.312 2.733-.715-1.607-2.377-2.733-4.313-2.733C5.
15        1 3.75 3 5.765 3 8.25c0 7.22 9 12 9 12s9-4.78 9-12z"
16    />
17  </svg>
18</div>
```

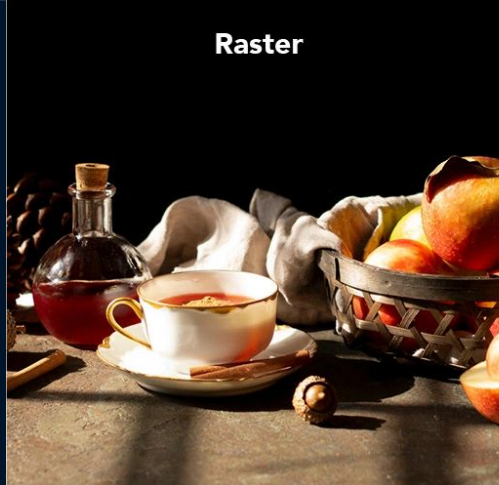


Layering of shapes goes from top to bottom in the code; the last element is most visible.

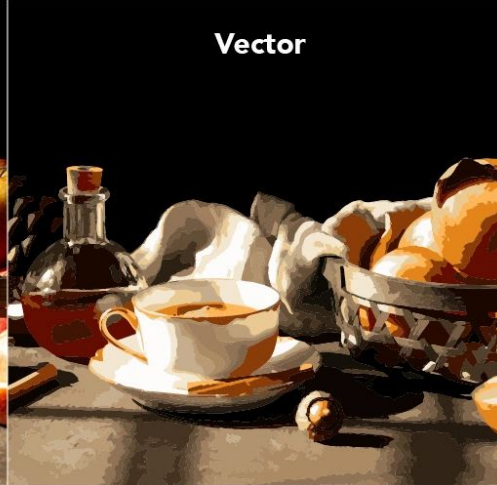
VS

Raster	Vector
Consist of pixels	Consist of mathematical formulas, shapes, and lines
Large file size	Very small file size
Scales with quality loss	Looks the same at any size
Can only be converted to other raster formats	Can be converted to raster images
Cannot be modified easily	Can be easily modified (e.g., splitting into components, animating)
Ideal for photographs	Ideal for icons

Raster

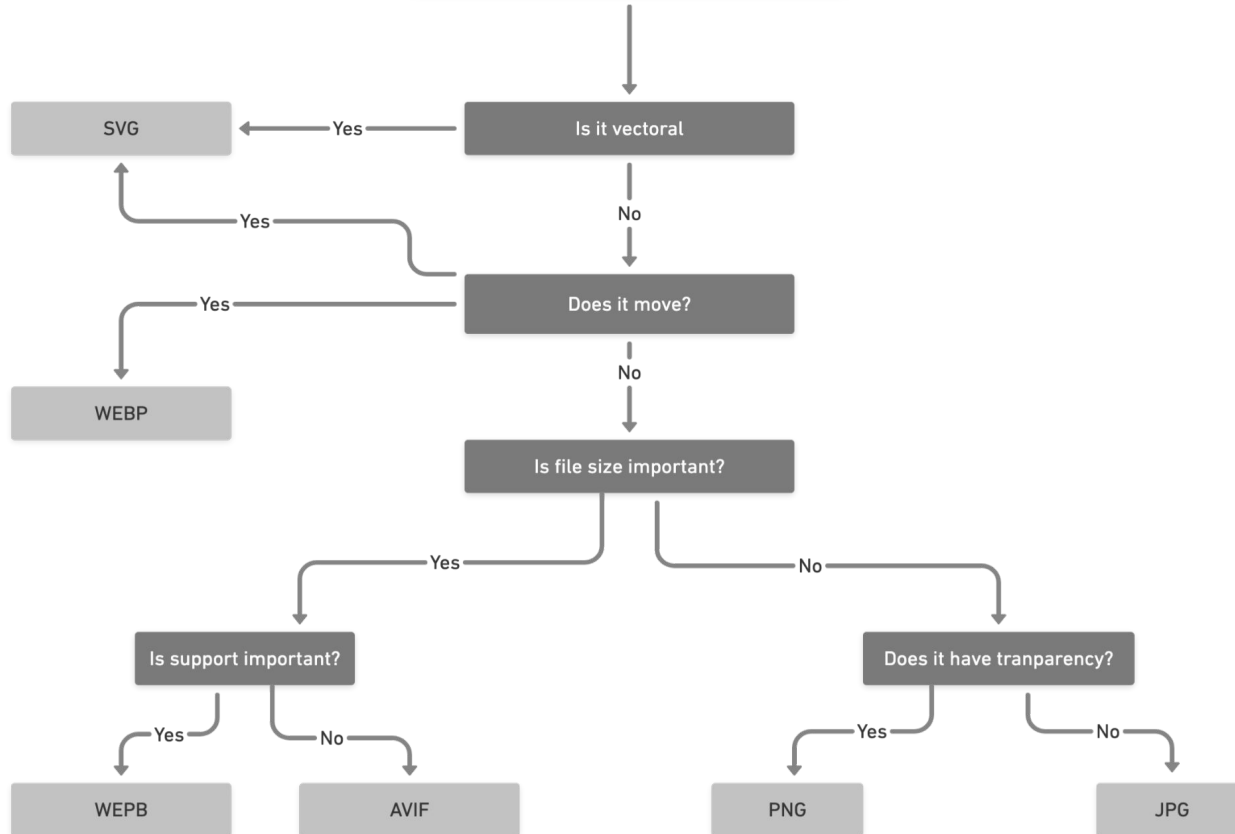


Vector



	JPEG Joint Photographic Experts Group	PNG Portable Network Graphics	GIF Graphics Interchange Format	WEBP	AVIF AV1 Still Image File Format
Lossless compression	Compression with quality loss			Better compression, retains quality	Excellent compression, retains quality
Size	Relatively small file size	Large file size		Weights 30-40% less than JPEG and PNG	Weights 10-20% less than WEBP
Detailisation	Common format for photos	Retains detail and color contrast		loss and blurring of fine details with strong compression	Suitable for large images, smaller images are better as WEBP
Text	hard to read text			may look blurry	
Supports transparency					
Colors		16 million colors	Limited colors (256)	Suitable for bright and colorful photos as it can display more pixels	Best choice for video, animations, and images with transparency
Animation					
Browser support	100%	100%	100%	96%	75%

What image format should you pick for your project?



N

Insertion

Methods of Inserting

Raster Images

- HTML

- `img`
- `picture`
- `figure`
- `figcaption`



```
<figure class="wiggly wiggle2">
  
  <figcaption>Wassily Kandinsky: Park of St Cloud,
1906</figcaption>
</figure>
```

- CSS

- `background-image`

```
#app #app-background #app-background-image { vYpXNJd?view
  background-image: url(
    https://images.unsplash.com/photo-14837286423... );
  height: 100%;
  transition: filter 250ms, transform 250ms;
  width: 100%;
}

.background-image { vYpXNJd?vie
  background-position: center;
  background-repeat: no-repeat;
  background-size: cover;
}
```



Wassily Kandinsky: Park of St
Cloud, 1906

Choosing the **Insertion** Method

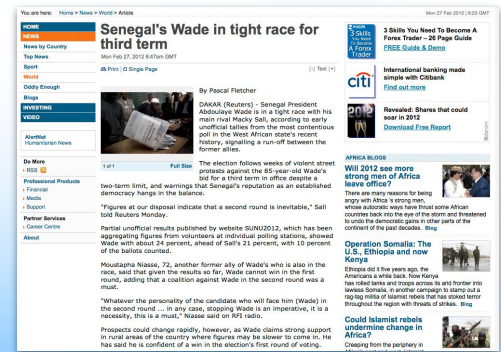
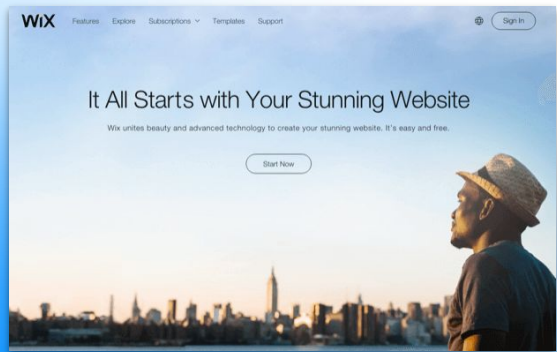
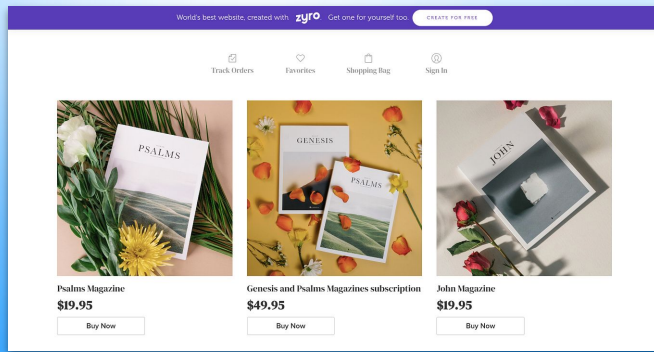
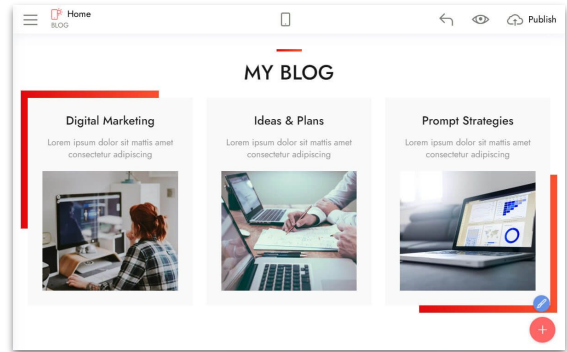
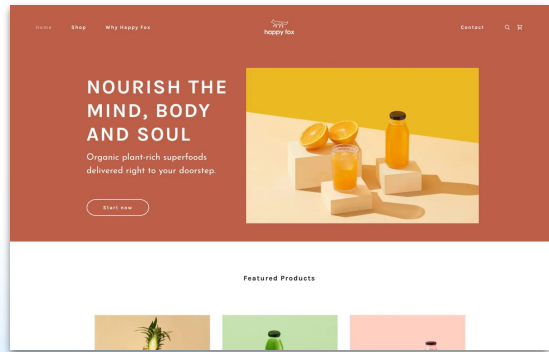
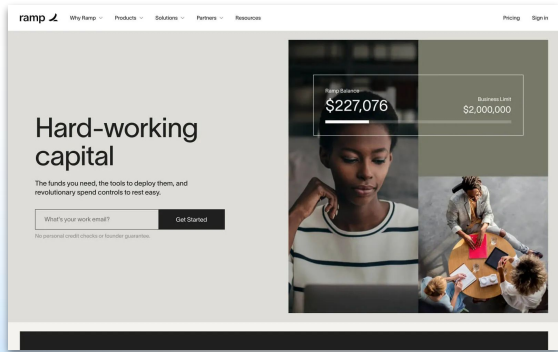
Content Images – `<img>` HTML

- Essential for user understanding
- Important for SEO
- Take up space in the document flow

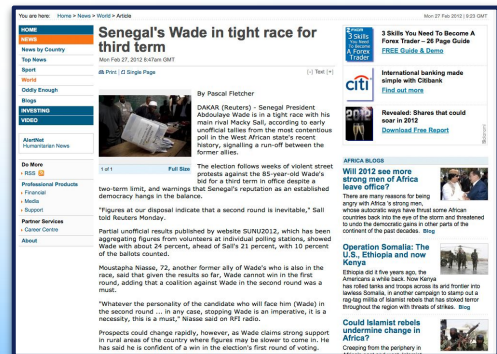
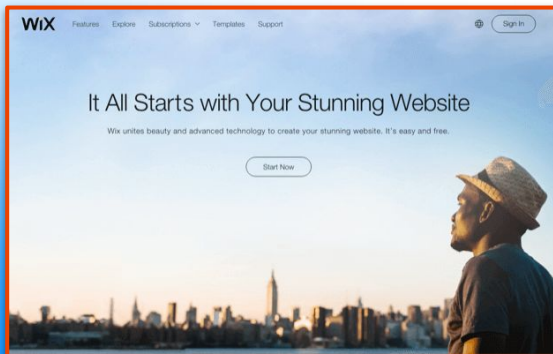
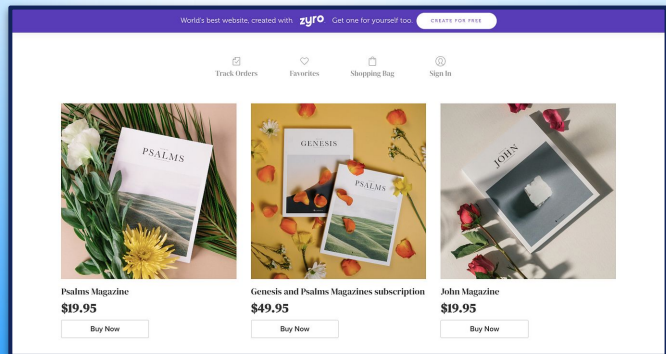
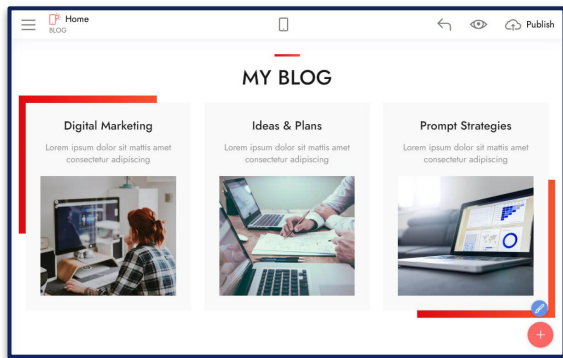
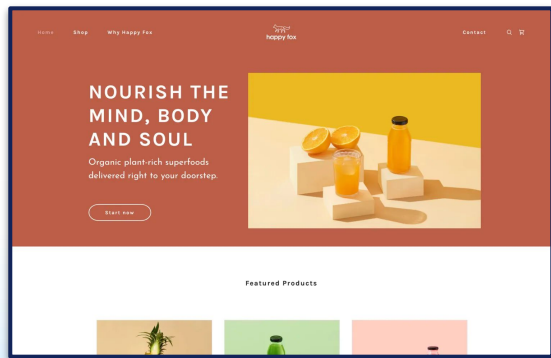
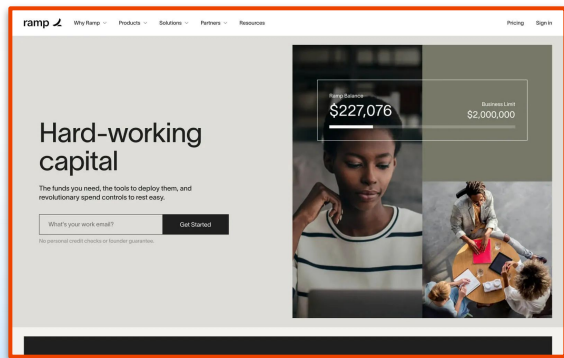
Decorative Images – `background-image` CSS

- Non-essential for user understanding
- Used for decoration
- Not important for SEO
- Often have other elements overlaid

content VS decorative



content VS decorative



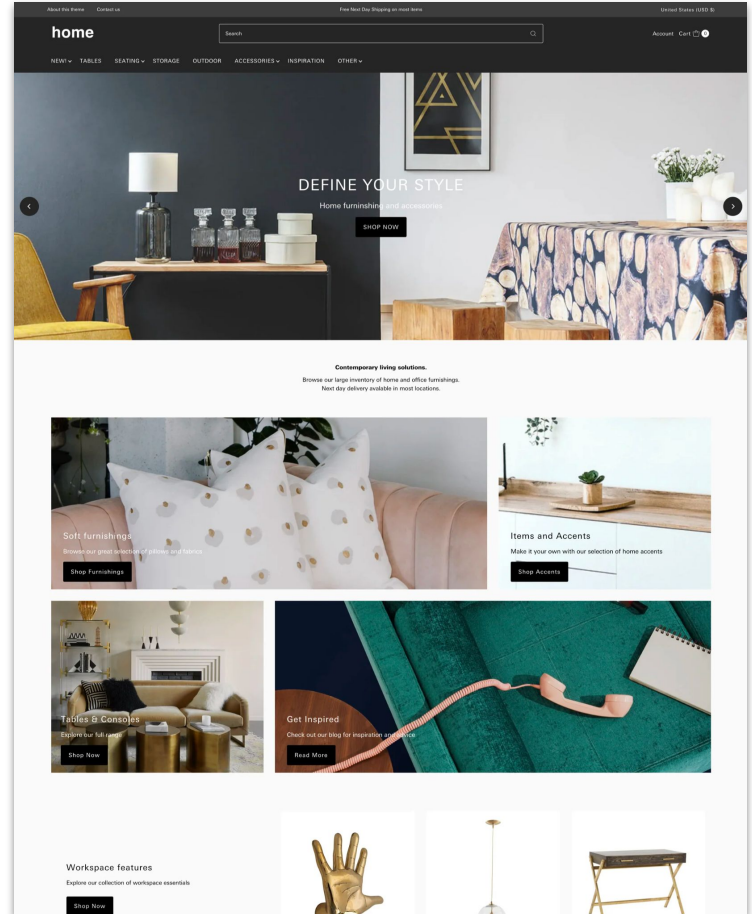
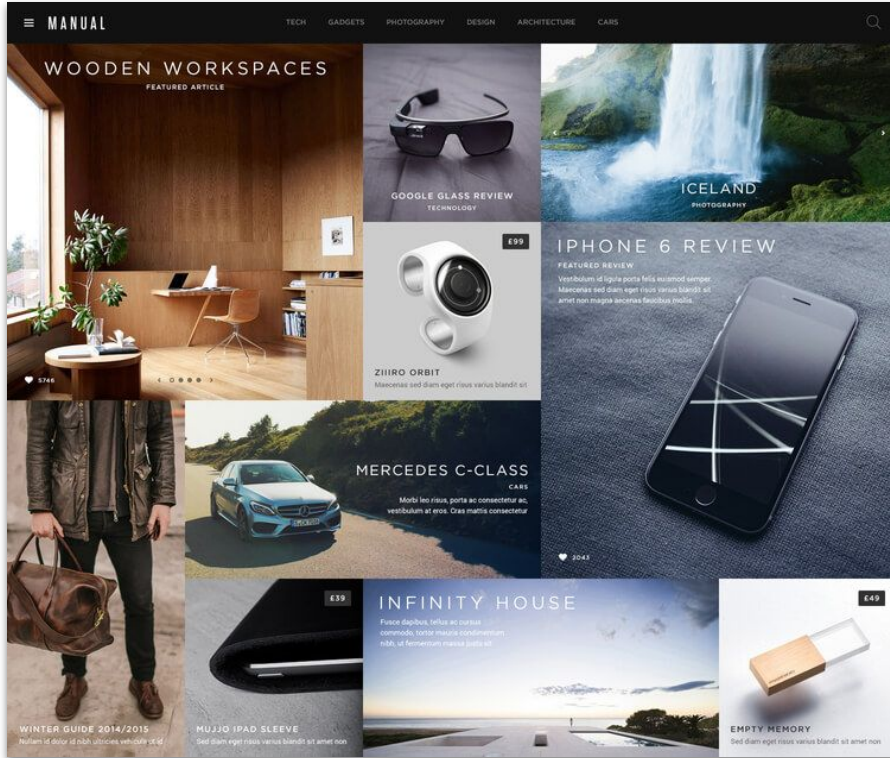


Image insertion



Inserting

Vector Images

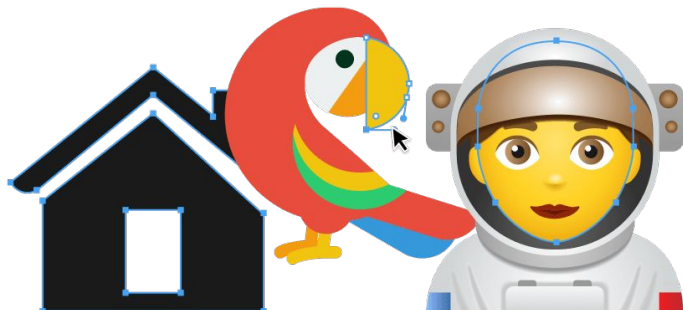
```
.block {  
  background-image: url(image.svg);  
  list-style-image: url(image.svg);  
  content: url(image.svg);  
}
```

```

```

```
<svg width="300" height="200"  
  xmlns="http://www.w3.org/2000/svg">  
  <rect width="100%" height="100%"  
    fill="red" />  
</svg>
```

```
<svg viewBox="0 0 30 10"  
  xmlns="http://www.w3.org/2000/svg">  
  <use xlink:href="image.svg" />  
</svg>
```





```

```

File naming:

- Use only English words
- No spaces
- No capital letters
- Reflect the location or subject of the image

```
blue triangle.jpg
BigPicture.jpg
```

cute-kitten.jpg
feature-block-1.jpg

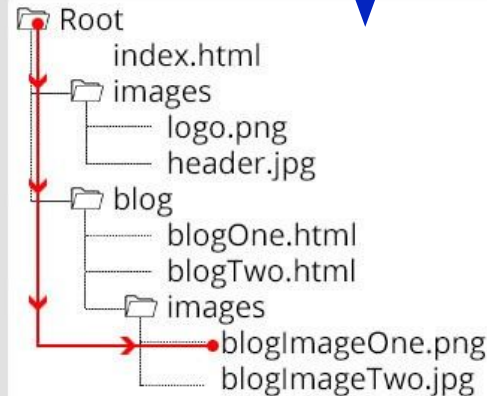
Paths

- **Absolute**
 - starts with slash
 - applicable to server
- **Relative**
 - starts with filename or dot and slash ./ or two dots and slash ../
 - applicable to file system

`src="https://MySite.com/blog/images/blogImageOne.png"`

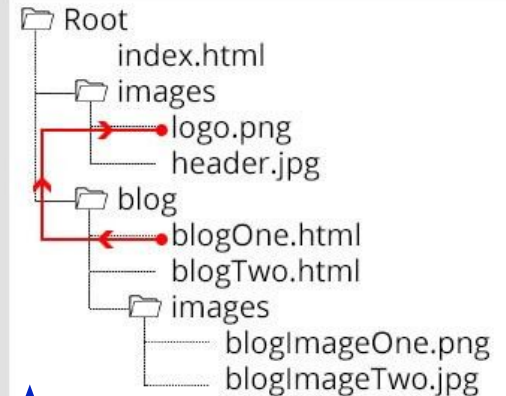
`src="/blog/images/blogImageOne.png"`

Absolute



`http://www.MySite.com/blog/images/blogImageOne.png`

Relative



`../images/logo.png`

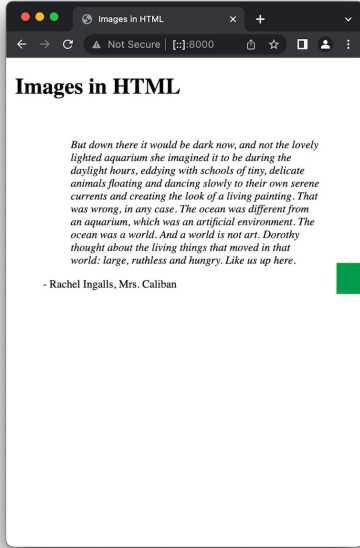
`src="../images/logo.png"`



```
width="100" height="100"
```

Attributes:

- **width/height**: Specifies dimensions of the image
- Used for displaying images before styles load, in reading mode, and for accessibility
- !without px

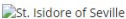


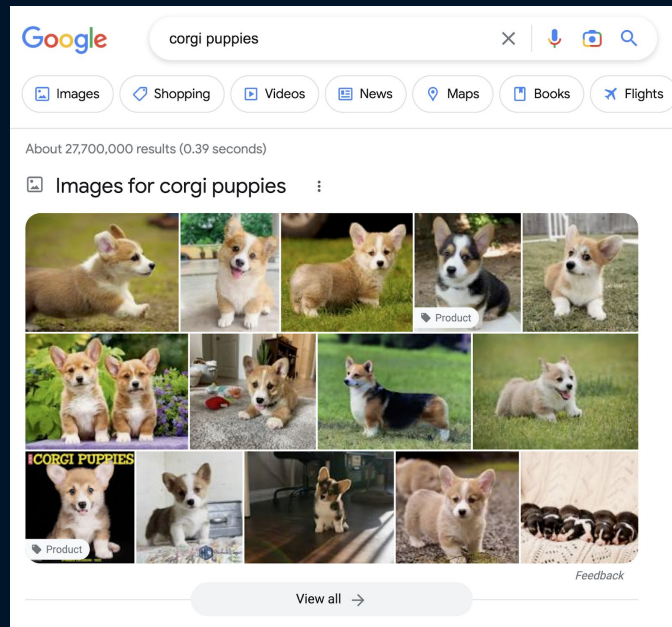
Importance of the **alt** Attribute

```

```

- Screen readers
- Search engines
- Broken images

Digital image that loaded correctly	Digital image that did not load correctly and does not have alt text	Digital image that did not load correctly and does have alt text
		

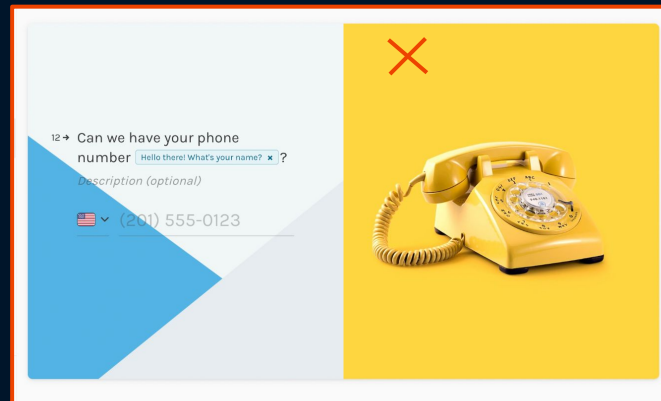
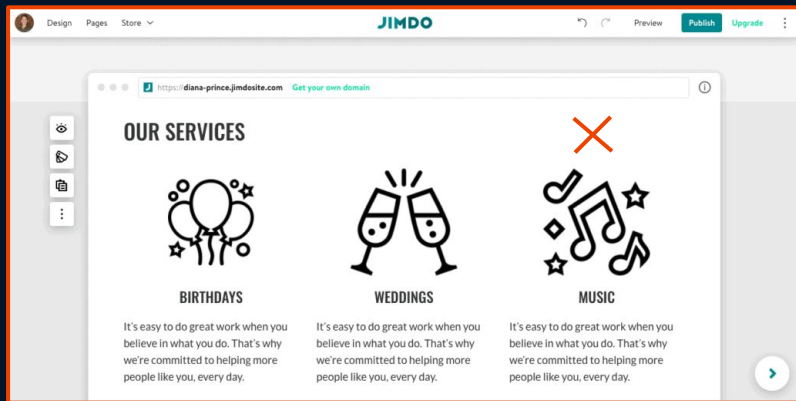


When **alt** is not needed

```

```

- Decorative and stock images
- Icons
- When the image caption matches the **alt** text





- ✓ Man wearing a helmet riding a bicycle
- ✗ bicycle
- ✗ Image of someone riding a bicycle



- ✓ Lay's Classic Potato Chips, 1.5 Ounce
- ✗ Potato Chips
- ✗ buy potato chips online



- ✓ Free SEO mini course for beginners by reliablesoft academy.
- ✗ download now free
- ✗ register to get free course



Good alt Practices

- **Summarize** what's in the image
- **Describe** as accurately as possible
- **Avoid declaration** like "image", "photo", "picture"
- Use the **context** of the surrounding content
- **Avoid repeating** text that is in the content
- **Don't use very long descriptions**, use other attributes for detailed descriptions
- Write in the **site's language**

Ask AI



help me to write alt text for this image in english

Thought for 8s

Here are concise, ready-to-use alt texts (in order):

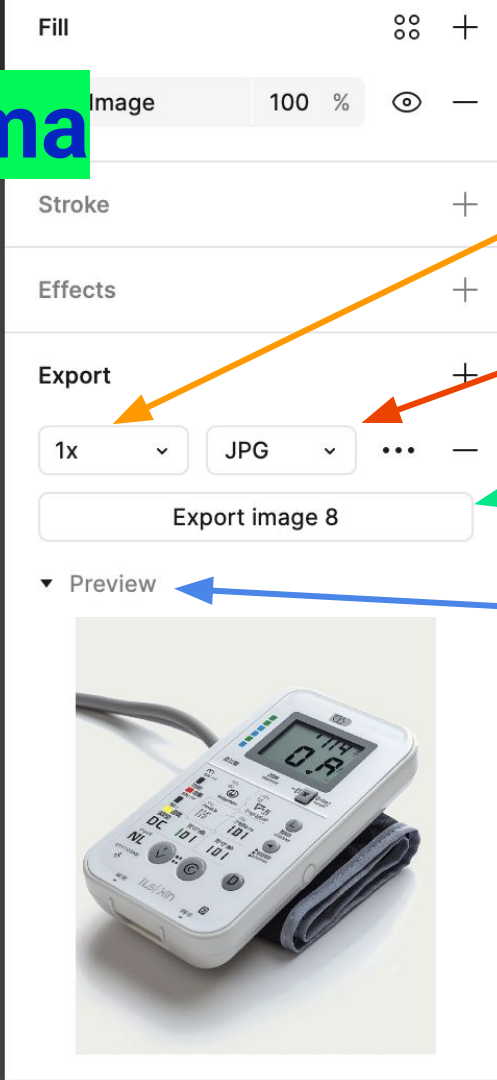
1. `alt="Giant green dragon confronts a cloaked figure in moss-lit ruins."`
2. `alt="Close-up of an emerald dragon's head with glittering scales and green bokeh."`
3. `alt="Enormous green dragon lowers its head toward a lone swordsman in rainy ruins."`

If any are purely decorative, use `alt=""` so screen readers skip them.

Export from Figma



461 x 558



scale factor (choose 2x or 3x)

image format

Export button

image preview

Optimize



Why Optimize Images

conversion (%)



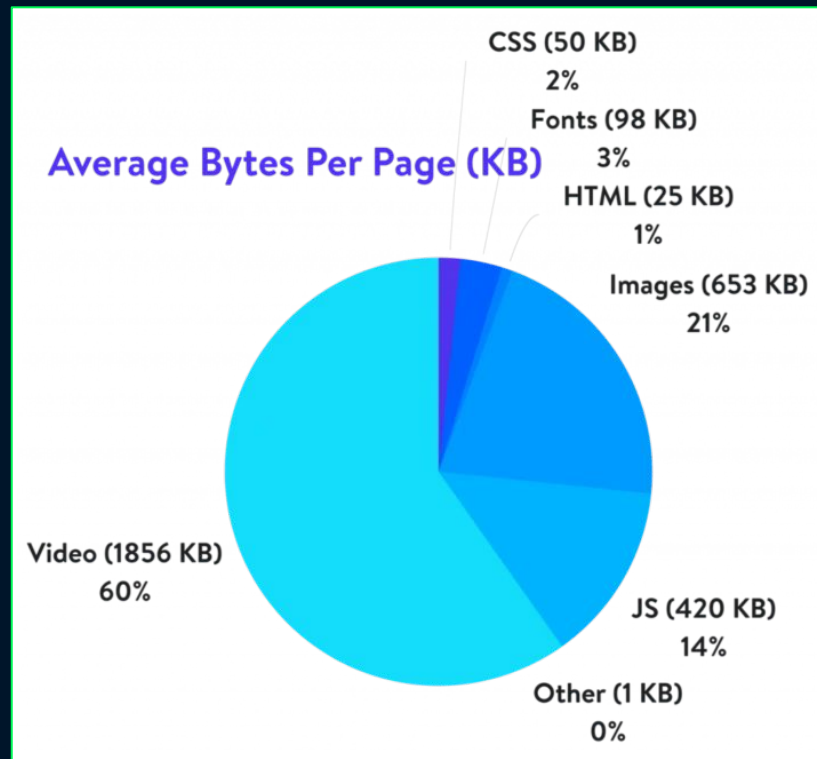
loading time (sec)

- Improves website loading speed
- Saves traffic and server resources
- Enhances website ranking in search results
- Increases conversion rates

Optimization

Methods

- Defer offscreen images
- Reduce file size
- Crop to exact size
- Change format
- Use CSS styling instead of image merging
- Decomposition
- Using patterns



Optimization methods

Defer offscreen images

```

```

load:

eager tells the browser to load the image as soon as the `<img>` element is processed

lazy tells the browser to load the image shortly before it will appear in the viewport

decoding:

async tells the browser to decode the image asynchronously and allow other content to be rendered before this completes

sync tells the browser to decode the image synchronously for atomic presentation with other content.

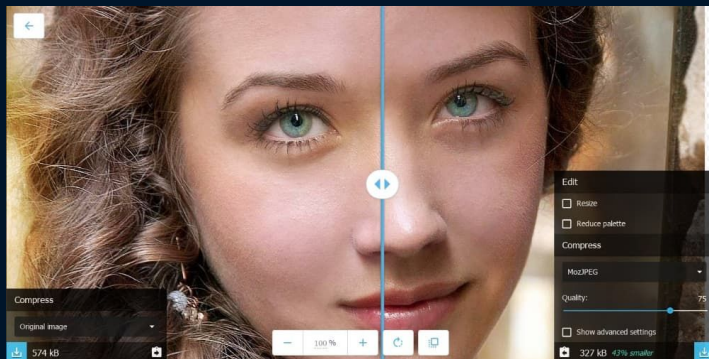
auto tells the browser to decide how to decode

Optimization methods

Reduce File Size

For example

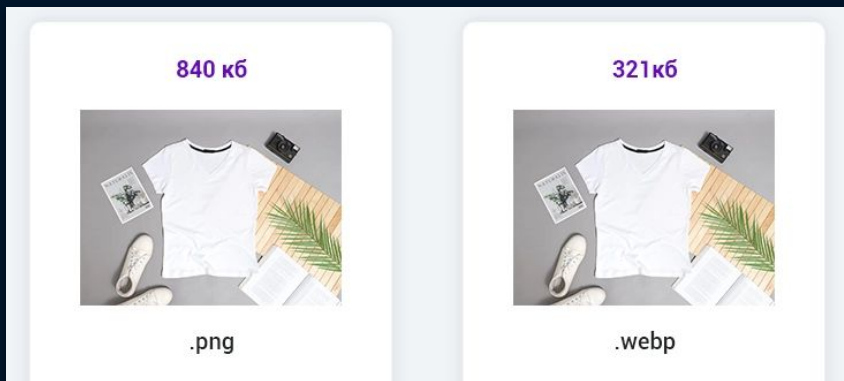
- [squoosh.app](#)
- [SVGOMG](#)



Optimization methods

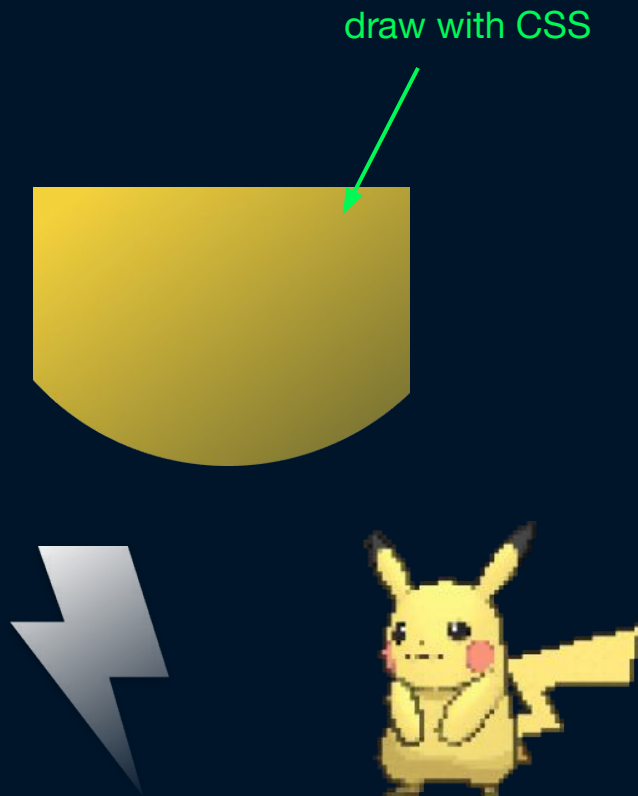
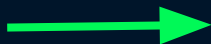
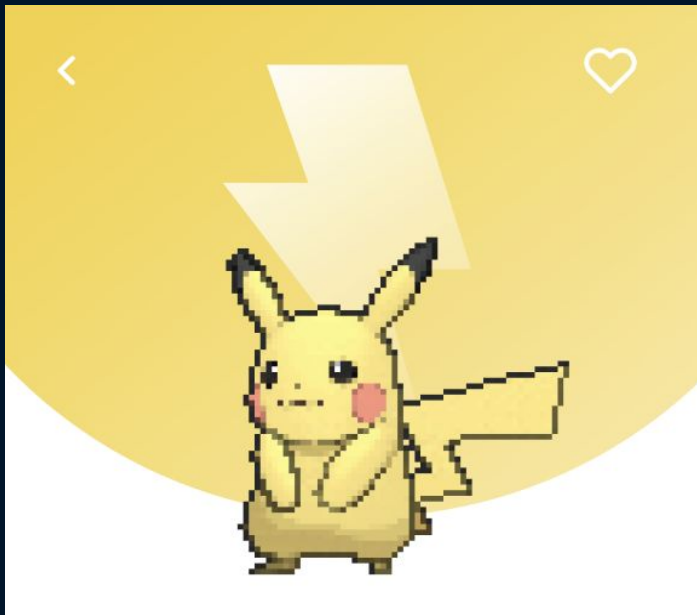
Change format

- Choose a more modern format
- Convert raster to vector



Optimization methods

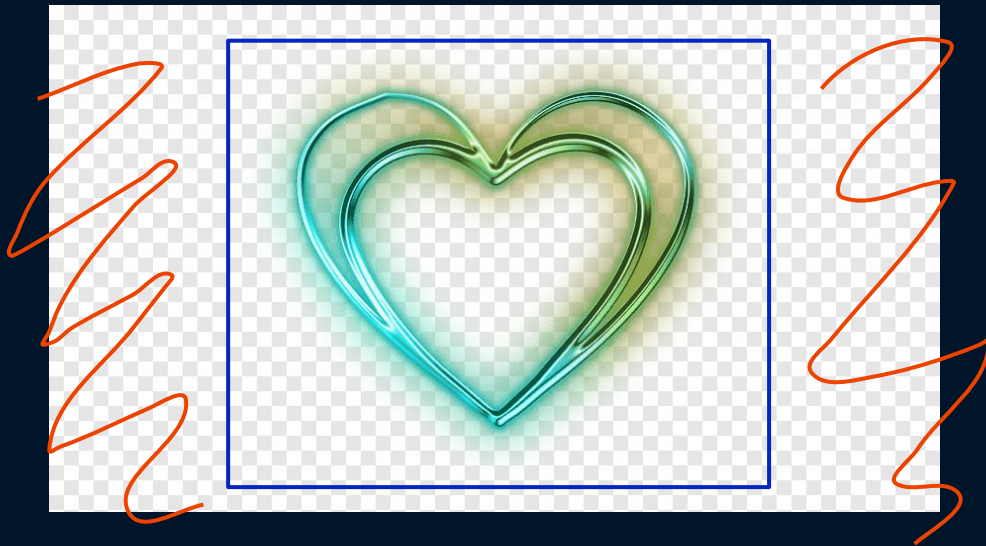
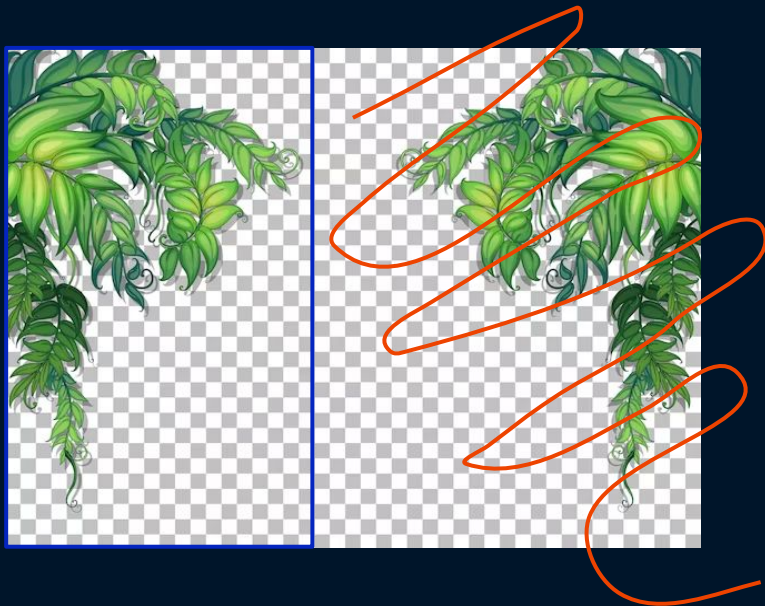
Decomposition



Optimization methods

Crop to exact size

No empty areas or unnecessary transparency

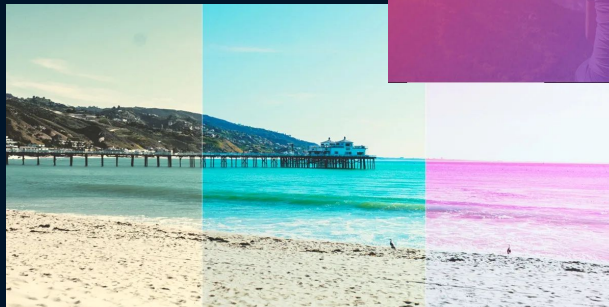
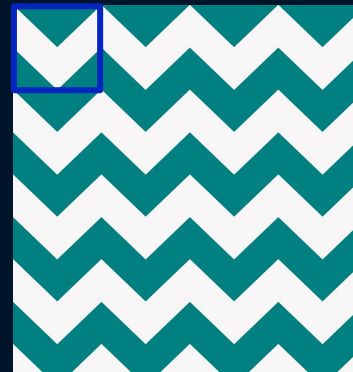


Optimization Methods

- Use CSS Styling Instead of Image Merging:

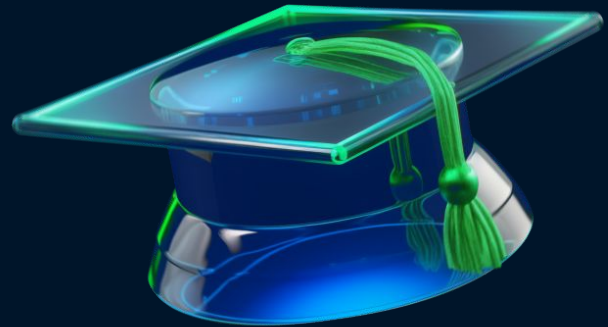
- Borders
- Gradients
- Filters
- Transformations (mirror, rotate, tilt)

- Using patterns



Summary

1. Difference between image formats
2. Difference between **raster and vector**
3. Choosing the right **image format**
4. Selecting the **insertion method** for images
5. How and why to **optimize images**



Quality Criteria for HTML Course

❤️ Mandatory for passing the course

💛 Required for the highest grade

💚 Optional

❤️ Correct Image Format

❤️ Extract images without style filters from the design.

❤️ Correct Image Insertion method

❤️ Specify dimensions for all images

❤️ Provide alt text for all images.

💛 Optimize images for browser rendering.

💛 Elements with background images should have a matching background color.

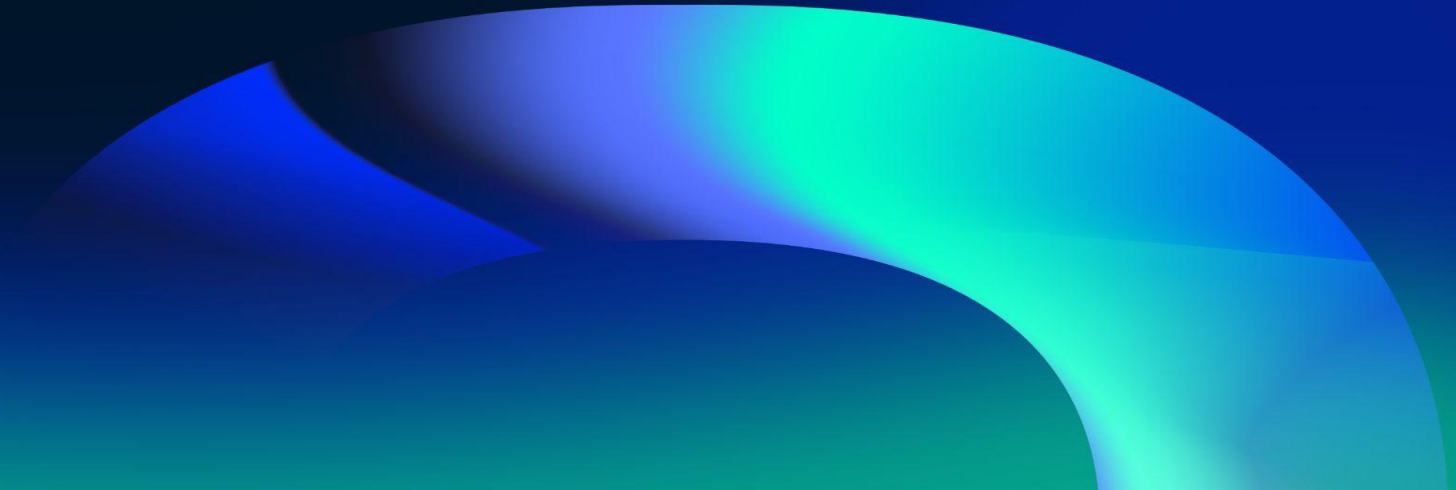
💚 Favicon icons display correctly.

B Academy
RO



QUESTIONS?

Please fill out the feedback form
It's very important for us





THANK YOU!

Have a good evening!